COCOMO Model – Formulas + Exam Solutions

Basic • Intermediate • Detailed (Examples 4.2, 4.3, 4.4)

1. Basic COCOMO – Formulas

Effort (E) = a × $(KLOC)^b$ Person-Months (PM)

Time (T) = c × $(E)^d$ Months

Cost = T × Salary per month (as done in the slide/book)

Project type	a	b	c	d
Organic (small, simple, in-house)	3.2 (slide)*	1.05	2.5	0.38
Semi-detached (medium)	3.0	1.12	2.5	0.35
Embedded (complex, hardware-tied)	2.8	1.20	2.5	0.32

*Original Boehm Basic COCOMO uses a = 2.4 (organic), 3.0, 3.6. Your slide uses 3.2 for organic, so use the constants your teacher/book gives. The solved examples below follow the slide (3.2).

2. Intermediate COCOMO – Formulas

Ei (initial effort) = a × $(KLOC)^b$

EAF = product of all cost-driver multipliers (C1 × C2 × ... × Cn)

E = Ei × EAF **T** = 2.5 × $(E)^{0.38}$

Rating words → multiplier: *very important / very high / highly considerable* = multiplier > 1 (extra effort); *low* = multiplier < 1; *nominal* = 1.00 (so skip it). In the question, the multiplier values for each cost driver come from the COCOMO cost-driver table; in Example 4.3 they are 1.16, 0.82, 0.91, 1.06, 1.10, 0.87.

3. Detailed COCOMO – Phase-wise distribution

Percent for a size between two table rows (linear interpolation):

% = P1 + [(P2 – P1) / (S2 – S1)] × (S – S1)

P1, P2 = % at the lower / upper size row; S1, S2 = those sizes (KLOC); S = your project size.

Phase effort = % × E **Phase time** = % × T

Table 4.4 – Organic rows only (needed for 4.4)

Organic	Req. & product design	Detailed design	Code & unit test	Integration & test
Effort % (2 KLOC)	16	26	42	16
Effort % (32 KLOC)	16	24	38	22
Time % (2 KLOC)	19	24	39	18
Time % (32 KLOC)	19	21	34	26

Semi-detached: 32 / 128 KLOC rows | Embedded: 128 / 320 KLOC rows (see Table 4.4 in your slide).

Calculator tip for x^y

Use the x^y key. Or: $x^y = e^{y \times \ln(x)}$. Example: $10^{1.05} = 11.22$ → 3.2 × 11.22 = 35.90. In the exam always write the substituted formula first, then the answer with unit (PM / Months / Rs).

Example 4.2 – Basic COCOMO

Given: Size = 10,000 LOC = **10 KLOC**; Salary = Rs. 25,000 / month. **Find:** Effort, Time, Total cost.

Step 1 – Identify project type

Simple student registration system, small size, in-house, simple → **Organic**. So a = 3.2, b = 1.05, c = 2.5, d = 0.38.

Step 2 – Effort

$E = a \times (KLOC)^b$
$E = 3.2 \times (10)^{1.05}$
$E = 3.2 \times 11.22$
E = 35.90 PM

Step 3 – Development time

$T = c \times (E)^d$
$T = 2.5 \times (35.90)^{0.38}$
$T = 2.5 \times 3.899$
T = 9.747 Months

Step 4 – Total cost

Cost = T × Salary
Cost = 9.747 × 25,000
Cost = Rs. 2,43,675

Note: the slide/book computes cost as Time × Salary, so write it this way in your exam. (Strictly, effort × salary = 35.90 × 25,000 = Rs. 8,97,500 is the 'per person' view; use it only if your teacher says so.)

Final Answer: Effort = 35.90 PM, Time = 9.747 months, Cost = Rs. 2,43,675.

Example 4.3 – Intermediate COCOMO

Given: 5 components: Online data entry 1.0, Data update 2.0, File I/O 1.5, Library reports 2.0, Query & search 0.5 (KLOC). Cost drivers given (rest nominal). **Find:** Effort and Time.

Step 1 – Total size

Size = 1.0 + 2.0 + 1.5 + 2.0 + 0.5 = **7 KLOC** → small, academic → **Organic** (a = 3.2, b = 1.05).

Step 2 – Initial effort

$Ei = 3.2 \times (7)^{1.05} = 3.2 \times 7.713$

Ei = 24.689 PM

Step 3 – EAF (only the drivers mentioned; nominal = 1)

EAF = 1.16 × 0.82 × 0.91 × 1.06 × 1.10 × 0.87

EAF = 0.878

Step 4 – Total effort

E = Ei × EAF = 24.689 × 0.878

E = 21.677 PM

Step 5 – Development time

$T = 2.5 \times (E)^{0.38} = 2.5 \times (21.677)^{0.38} = 2.5 \times 3.219$

T = 8.047 Months

Final Answer: Effort = 21.677 PM, Time = 8.047 months.

Example 4.4 – Detailed COCOMO (continuing 4.3)

Given: E = 21.677 PM, T = 8.047 months, Size = 7 KLOC, Organic. 7 lies between 2 KLOC and 32 KLOC → interpolate. **Find:** phase-wise effort and time.

Interpolation: % = P1 + [(P2 – P1) / (32 – 2)] × (7 – 2)

The slide uses (S – S1) shown as 7 and rounds to the nearest whole %. Same values are used below so your answer matches the book.

(A) Phase-wise EFFORT (E = 21.677 PM)

Phase	Calculation of %	%	Effort = % × 21.677
Requirements & product design	16 + ((16–16)/30) × 7	16	0.16 × 21.677 = 3.468 PM
Detailed design	26 + ((24–26)/30) × 7	26	0.26 × 21.677 = 5.636 PM
Code & unit test	42 + ((38–42)/30) × 7	41	0.41 × 21.677 = 8.888 PM
Integration & test	16 + ((22–16)/30) × 7	17	0.17 × 21.677 = 3.685 PM

(B) Phase-wise TIME (T = 8.0467 months)

Phase	Calculation of %	%	Time = % × 8.0467
Requirements & product design	19 + ((19–19)/30) × 7	19	0.19 × 8.0467 = 1.529 M
Detailed design	24 + ((21–24)/30) × 7	23	0.23 × 8.0467 = 1.851 M
Code & unit test	39 + ((34–39)/30) × 7	38	0.38 × 8.0467 = 3.058 M
Integration & test	18 + ((26–18)/30) × 7	20	0.20 × 8.0467 = 1.609 M

Check: effort % total = 16+26+41+17 = 100; time % total = 19+23+38+20 = 100. Always add the percentages to confirm they make 100.

Exam tips: (1) Write: Given → Type of project → Formula → Substitution → Answer with unit. (2) Always state why the project is Organic / Semi-detached / Embedded. (3) Use PM for effort, Months for time, Rs for cost.